Echo-Loop: A Building That Runs Itself

Autonomous closed-loop HVAC control using EnergyPlus, MCP and a local open-source LLM.

EnergyPlus supplies the physics. A local open-source LLM supplies the judgement. The Model Context Protocol carries every reading and every command between them — while the simulation is still running.

−8.5%
HVAC Energy
Saved over a summer week

336 / 336
Agent Turns
Zero failures

100%
Open Source
One laptop — no cloud, no API keys

Problem Statement ID <<FILL IN>> · **Theme** <<FILL IN>>

PS Category Software · Student Ayush Yadav · Student ID <<FILL IN>>

Repository github.com/ayushYadav1107/Eco_loop
Live demo eco-loop-agents.streamlit.app

ENERGYPLUS → MCP TOOLS → LOCAL LLM → SAFETY GATE → LIVE ACTUATORS

one closed loop, every 60 simulated minutes

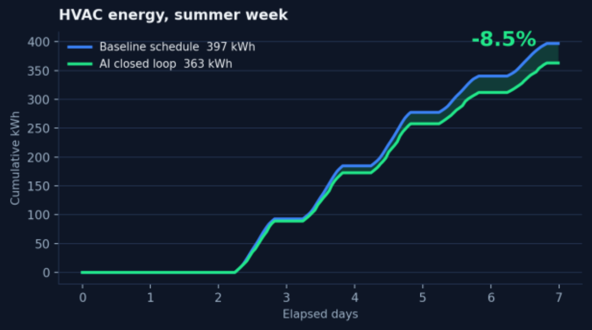
ENERGYPLUS 26.1 · PYENERGYPLUS · FASTMCP · MODEL CONTEXT PROTOCOL · OLLAMA + LLAMA 3.2 3B · PYDANTIC · STREAMLIT

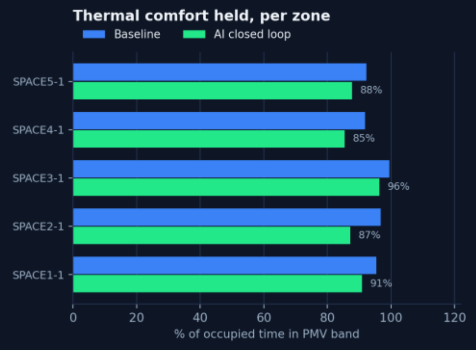
Traditional BMS follow rigid clock schedules.

Echo-Loop turns the building into a self-correcting agent.

How It Works

- EnergyPlus runs in-process; sensors read every zone timestep
- Every 60 simulated minutes the state reaches a local LLM via six MCP tools
- The model weighs comfort, demand and grid carbon, then returns setpoints





Why It's Different

Most LLM-and-simulation work rewrites the model file and re-runs it — it cannot react to a zone drifting out of comfort at 14:00 on day 3. Echo-Loop does.

ONE LINE FROM A LIVE RUN

```
[ai 07-17 09:00] OAT=24.9C PMV=-1.02 kW=2.09
   -> cool=24.8 heat=19.2 (llm, 1.3s)
```

Injected into the LIVE solver

No IDF rewrite, no restart.

Setpoints reach the actuator handles mid-solve.

Safety Boundary

MCP validates every command server-side.

Five gates; a rejected command never reaches the model.

Measured, Not Projected

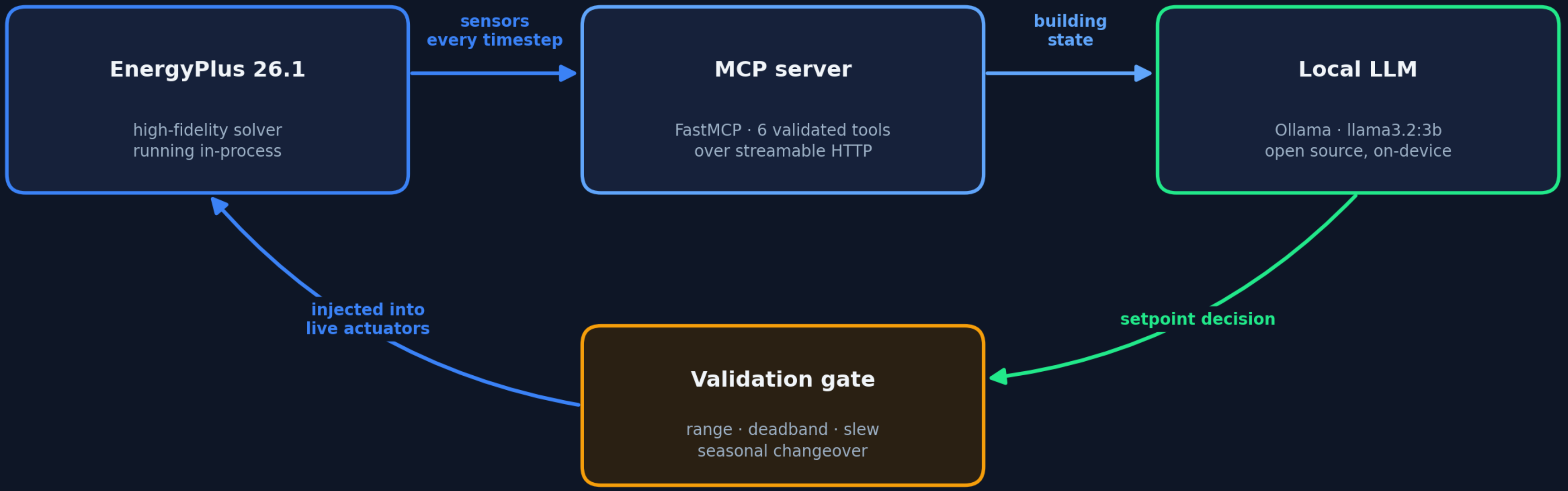
PMV −0.43 → −0.02 while saving 8.5%.

Both runs are committed EnergyPlus output.

TECHNICAL APPROACH

The Closed Loop Architecture

CLOSED LOOP · no file rewrite, no restart · 168 agent decisions per simulated week



ONE CONTROL INTERVAL · 60 SIMULATED MINUTES

the solver is stopped only for the agent turn

Sample → Aggregate → LLM tool-calling turn → Validate gate → Inject into live actuators via the pyenergyplus C API

Feasibility and Overcoming Constraints

336 / 336

Agent Turns

0 fallbacks · 0 timeouts · 0 errors

−3.7%

Winter Energy

Same loop, heating season

2.1 s

Median Latency

Fully on-device

Risks

- LLM latency inside a blocking simulation callback
- A small model can move the wrong setpoint, or run heating against cooling
- One setpoint pair cannot satisfy five differently-loaded zones

Solutions

- Hard per-turn deadline; on timeout the loop holds the last accepted command
- Observations state the required direction — the model cannot invert physics
- Seasonal changeover parks the idle setpoint; a per-zone trim corrects each zone

EVERY COMMAND PASSES FIVE GATES BEFORE IT REACHES AN ACTUATOR

1 SCHEMA range · deadband → **2 CHANGEOVER** park the idle mode → **3 SLEW LIMIT** ≤ 3 °C / hour → **4 PER-ZONE TRIM** each zone's own PMV → **5 FALLBACK** hold last good command

Rejected commands return accepted: false with a reason — the previous setpoints stay in force and the model can retry.

Artifacts & References

Repository github.com/ayushYadav1107/Eco_loop **Live demo** eco-loop-agents.streamlit.app

Baseline and runtime-generated .idf models, agent_decisions.jsonl and per-timestep EnergyPlus results are all committed — a fresh clone reproduces every figure in this deck.

Simulation & Building Physics

- EnergyPlus Engineering Reference & EMS Application Guide
- pyenergyplus Data Exchange API — get_variable_handle, set_actuator_value
- ISO 7730:2005 & ASHRAE Standard 55 — PMV/PPD comfort model

Agent, Protocol & Inference

- Model Context Protocol — modelcontextprotocol.io
- FastMCP — gofastmcp.com
- Meta Llama 3.2 served locally through Ollama — ollama.com

168 setpoint decisions, one summer week

The agent's own stated reasoning, taken verbatim from agent_decisions.jsonl

07-17 09:00 *"Raise cooling_sp to about 27.0 C this turn — this fixes comfort AND saves energy"*

07-17 11:00 *"HOLD the current setpoints. Do not trim energy here — you would push occupants out of the band"*

07-17 12:00 *"Nudge cooling_sp up to save energy while keeping PMV within the band"*

Every decision is logged with the model's own reason

outputs/ai/agent_decisions.jsonl — committed, one JSON record per control interval.

Auditable, Not Anecdotal

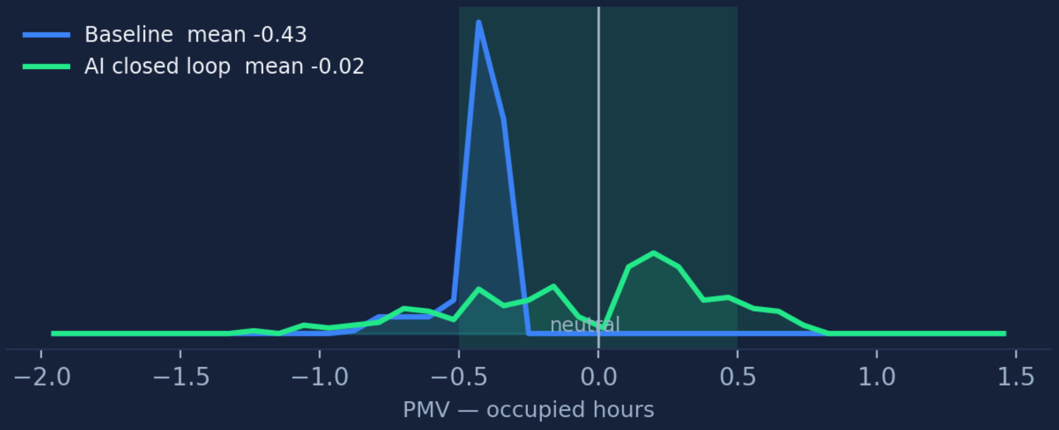
The controller runs at temperature 0 — repeated runs return identical totals. Anyone can clone the repo and reproduce these exact figures.

```
python main.py prepare
python main.py run-baseline --start 07-15 --end 07-21
python main.py run-ai --start 07-15 --end 07-21
python main.py dashboard
# ~6 min end to end on a laptop — EnergyPlus 26.1 + Ollama,
# no cloud, no API key, no paid service anywhere in the loop
```

Four Commands. Full Reproduction.

- prepare — stage and instrument the EnergyPlus model
- run-baseline — the same building on its native schedule
- run-ai — the closed loop, 168 agent decisions
- dashboard — the savings comparison, side by side

Comfort was not traded away



Honest limitation

One setpoint pair drives five differently-loaded zones. Per zone the agent holds 85–96% of occupied time inside the PMV band; the stricter worst-zone metric is lower.

Per-zone setpoints are the next step — the actuator handles and per-zone PMV are already in place.

Stated in full in ARCHITECTURE.md § 8 — nothing here is hidden.